

DBS LAB 9

IN – LAB TASK

Attributes:

Name

Course_ID

	Name	Data type
1	Course_ID	Integer
2	Course_Name	VARCHAR (20)
3	Department	VARCHAR (50)
4	Instructor_ID	VARCHAR

Attribute Properties

Name: Course_ID

Data Type: ☐ Domain ☒ Logical ☐ Distinct
☐ Structured ☐ Collection

Source Type: Integer Preferred ☐

☒ Primary UID ☐ Relation UID ☒ Mandatory ☐ Deprecated

Details Overview UDP

Attributes:

Name

Student_id

	Name	Data type
1	Student_id	Integer
2	First_Name	VARCHAR (20)
3	Last_Name	VARCHAR (20)
4	E-mail	VARCHAR (100)
5	Registration_Date	Date

Attribute Properties

Name: Student_id

Data Type: ☐ Domain ☒ Logical ☐ Distinct
☐ Structured ☐ Collection

Source Type: Integer Preferred ☐

☒ Primary UID ☐ Relation UID ☒ Mandatory ☐ Deprecated

Details Overview UDP

Attributes:

Name

Registration_ID

	Name	Data type
1	Registration_ID	Integer
2	Student_ID	Integer
3	Course_ID	Integer
4	Registration	Date

Attribute Properties

Name: Registration_ID

Data Type: ☐ Domain ☒ Logical ☐ Distinct
☐ Structured ☐ Collection

Source Type: Integer Preferred ☐

☒ Primary UID ☐ Relation UID ☒ Mandatory ☐ Deprecated

Details
Overview
UDP

Attributes:

Name

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	Name	Data type
1	Exam_ID	Integer
2	Course_ID	Integer
3	Exam_Type	VARCHAR (20)
4	Exam_Date	Date
5	Start_Time	Time
6	End_Time	Time
7	location	VARCHAR (100)

Attribute Properties

Name

Exam_ID

Data Type

☐ Domain
☒ Logical
☐ Distinct

☐ Structured
☐ Collection

Source Type

Integer

Preferred ☐

☒ Primary UID
☐ Relation UID
☒ Mandatory
☐ Deprecated

Details
Overview
UDP

Attributes:

Name

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	Name	Data type
1	Enrollment_ID	Integer
2	Student_ID	Integer
3	Exam_ID	Integer
4	Attendance	Integer

Attribute Properties

Name

Enrollment_ID

Data Type

☐ Domain
☒ Logical
☐ Distinct

☐ Structured
☐ Collection

Source Type

Integer

Preferred ☐

☒ Primary UID
☐ Relation UID
☒ Mandatory
☐ Deprecated

Attributes:

Name

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	Name	Data type
1	Make_Up_ID	Integer
2	Student_ID	Integer
3	Exam_ID	Integer
4	Make_Up_date	Date
5	Reason	VARCHAR (100)

Attribute Properties

Name

Make_Up_ID

Data Type

☐ Domain
☒ Logical
☐ Distinct

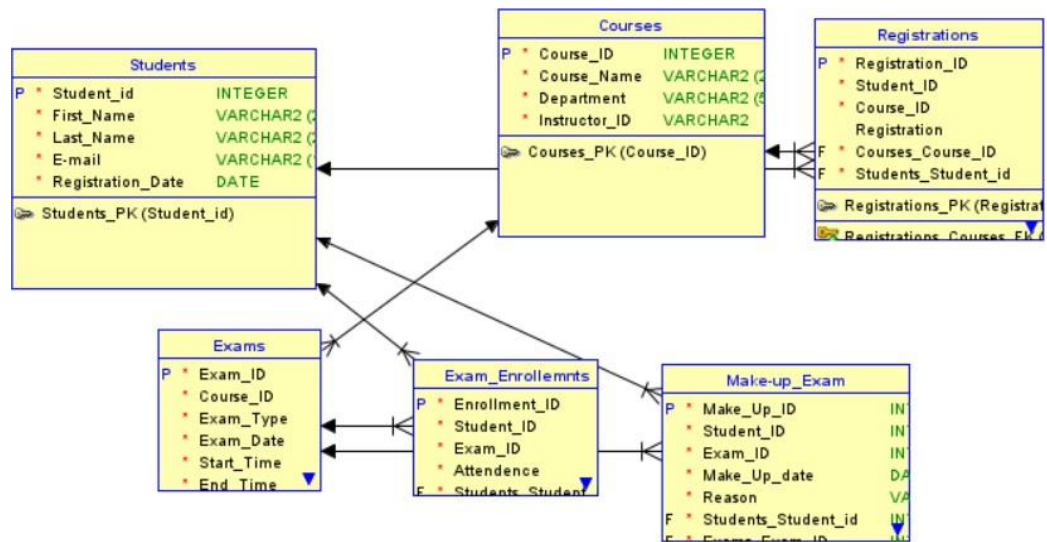
☐ Structured
☐ Collection

Source Type

Integer

Preferred ☐

☒ Primary UID
☐ Relation UID
☒ Mandatory
☐ Deprecated



```

-- Generated by Oracle SQL Developer Data Modeler 24.3.1.351.0831
--   at:      2026-04-04 20:16:13 PKT
--   site:    Oracle Database 11g
--   type:    Oracle Database 11g

-- predefined type, no DDL - MDSYS.SDO_GEOMETRY

-- predefined type, no DDL - XMLTYPE

CREATE TABLE Courses
(
  Course_ID      INTEGER NOT NULL ,
  Course_Name    VARCHAR2 (20) NOT NULL ,
  Department     VARCHAR2 (50) NOT NULL ,
  Instructor_ID  VARCHAR2
-- ERROR: VARCHAR2 size not specified
                NOT NULL
)
;

ALTER TABLE Courses
  ADD CONSTRAINT Courses_PK PRIMARY KEY ( Course_ID ) ;

CREATE TABLE Exam_Enrollemnts
(
  Enrollment_ID      INTEGER NOT NULL ,
  Student_ID         INTEGER NOT NULL ,
  Exam_ID            INTEGER NOT NULL ,
  Attendance         INTEGER NOT NULL ,
  Students_Student_id INTEGER NOT NULL ,
  Exams_Exam_ID      INTEGER NOT NULL
)
;

ALTER TABLE Exam_Enrollemnts
  ADD CONSTRAINT Exam_Enrollemnts_PK PRIMARY KEY ( Enrollment_ID ) ;

CREATE TABLE Exams
(
  Exam_ID          INTEGER NOT NULL ,
  Course_ID        INTEGER NOT NULL ,

```

CREATE TABLE Exams

```
(
  Exam_ID          INTEGER NOT NULL ,
  Course_ID        INTEGER NOT NULL ,
  Exam_Type        VARCHAR2 (20) NOT NULL ,
  Exam_Date        DATE NOT NULL ,
  Start_Time       DATE NOT NULL ,
  End_Time         DATE NOT NULL ,
  location         VARCHAR2 (100) NOT NULL ,
  Courses_Course_ID INTEGER NOT NULL
)
```

;

ALTER TABLE Exams

ADD CONSTRAINT Exams_PK PRIMARY KEY (Exam_ID) ;

CREATE TABLE "Make-up_Exam"

```
(
  Make_Up_ID       INTEGER NOT NULL ,
  Student_ID       INTEGER NOT NULL ,
  Exam_ID          INTEGER NOT NULL ,
  Make_Up_date     DATE NOT NULL ,
  Reason           VARCHAR2 (100) NOT NULL ,
  Students_Student_id INTEGER NOT NULL ,
  Exams_Exam_ID    INTEGER NOT NULL
)
```

;

ALTER TABLE "Make-up_Exam"

ADD CONSTRAINT "Make-up_Exam_PK" PRIMARY KEY (Make_Up_ID) ;

CREATE TABLE Registrations

```
(
  Registration_ID   INTEGER NOT NULL ,
  Student_ID       INTEGER NOT NULL ,
  Course_ID        INTEGER NOT NULL ,
  Registration      DATE ,
  Courses_Course_ID INTEGER NOT NULL ,
  Students_Student_id INTEGER NOT NULL
)
```

;

```
ALTER TABLE Registrations
  ADD CONSTRAINT Registrations_PK PRIMARY KEY ( Registration_ID )
```

⊞ CREATE TABLE Students

```
(
  Student_id          INTEGER NOT NULL ,
  First_Name          VARCHAR2 (20) NOT NULL ,
  Last_Name           VARCHAR2 (20) NOT NULL ,
  "E-mail"            VARCHAR2 (100) NOT NULL ,
  Registration_Date    DATE NOT NULL
)
```

```
;

ALTER TABLE Students
  ADD CONSTRAINT Students_PK PRIMARY KEY ( Student_id ) ;
```

⊞ ALTER TABLE Exam_Enrollemnts

```
  ADD CONSTRAINT Exam_Enrollemnts_Exams_FK FOREIGN KEY
  (
    Exams_Exam_ID
  )
  REFERENCES Exams
  (
    Exam_ID
  )
;
```

⊞ ALTER TABLE Exam_Enrollemnts

```
  ADD CONSTRAINT Exam_Enrollemnts_Students_FK FOREIGN KEY
  (
    Students_Student_id
  )
  REFERENCES Students
  (
    Student_id
  )
;
```



```

) ALTER TABLE Exams
  ADD CONSTRAINT Exams_Courses_FK FOREIGN KEY
  (
    Courses_Course_ID
  )
  REFERENCES Courses
  (
    Course_ID
  )
;

) ALTER TABLE "Make-up_Exam"
  ADD CONSTRAINT "Make-up_Exam_Exams_FK" FOREIGN KEY
  (
    Exams_Exam_ID
  )
  REFERENCES Exams
  (
    Exam_ID
  )
;

) ALTER TABLE "Make-up_Exam"
  ADD CONSTRAINT "Make-up_Exam_Students_FK" FOREIGN KEY
  (
    Students_Student_id
  )
  REFERENCES Students
  (
    Student_id
  )
;

) ALTER TABLE Registrations
  ADD CONSTRAINT Registrations_Courses_FK FOREIGN KEY
  (
    Courses_Course_ID
  )
  REFERENCES Courses
  (
    Course_ID
  )
;

```

```

ALTER TABLE Registrations
  ADD CONSTRAINT Registrations_Students_FK FOREIGN KEY
    (
      Students_Student_id
    )
  REFERENCES Students
    (
      Student_id
    )
;

```

```

-- Oracle SQL Developer Data Modeler Summary Report:

```

```

--
-- CREATE TABLE                                6
-- CREATE INDEX                                0
-- ALTER TABLE                               13
-- CREATE VIEW                                0
-- ALTER VIEW                                0
-- CREATE PACKAGE                             0
-- CREATE PACKAGE BODY                        0
-- CREATE PROCEDURE                           0
-- CREATE FUNCTION                           0
-- CREATE TRIGGER                             0
-- ALTER TRIGGER                             0
-- CREATE COLLECTION TYPE                     0
-- CREATE STRUCTURED TYPE                     0
-- CREATE STRUCTURED TYPE BODY                0
-- CREATE CLUSTER                             0
-- CREATE CONTEXT                             0
-- CREATE DATABASE                           0
-- CREATE DIMENSION                           0
-- CREATE DIRECTORY                           0
-- CREATE DISK GROUP                           0
-- CREATE ROLE                                0
-- CREATE ROLLBACK SEGMENT                     0
-- CREATE SEQUENCE                            0
-- CREATE MATERIALIZED VIEW                    0
-- CREATE MATERIALIZED VIEW LOG                0
-- CREATE SYNONYM                             0
-- CREATE TABLESPACE                         0

```


POST- LAB TASK

Details Overview UDP

Attributes:

Name

1 PlanID Integer

2 PlanName VARCHAR (20)

3 Description VARCHAR (100)

4 MonthlyRate DECIMAL (2, 10)

Attribute Properties

Name PlanID

Data Type ☐ Domain ☒ Logical ☐ Distinct

☐ Structured ☐ Collection

Source Type Integer Preferred ☐

☒ Primary UID ☐ Relation UID ☒ Mandatory ☐ Deprecated

Details Overview UDP

Attributes:

Name

1 ResidentID Integer

2 FullName VARCHAR (50)

3 HotelID Integer

4 RoomNumber Integer

5 Phone Integer

6 Attribute_6 Unknown

7 PlanID Integer

Attribute Properties

Name ResidentID

Data Type ☐ Domain ☒ Logical ☐ Distinct

☐ Structured ☐ Collection

Source Type Integer Preferred ☐

☒ Primary UID ☐ Relation UID ☒ Mandatory ☐ Deprecated

Details Overview UDP

Attributes:

Name

1 ProfileID Integer

2 ResidentID Integer

3 Preference VARCHAR (20)

4 Allergies VARCHAR (100)

Attribute Properties

Name ProfileID

Data Type ☐ Domain ☒ Logical ☐ Distinct

☐ Structured ☐ Collection

Source Type Integer Preferred ☐

☒ Primary UID ☐ Relation UID ☒ Mandatory ☐ Deprecated

Details
Overview
UDP

Attributes:

Name

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	Name	Data type
1	BookingID	Integer
2	ResidentID	Integer
3	BookingDate	Date
4	MealSlot	VARCHAR (20)
5	IsGuestBooking	Boolean (1)
6	Status	VARCHAR (20)

Attribute Properties

Name

BookingID

Data Type

☐ Domain
☒ Logical
☐ Distinct

☐ Structured
☐ Collection

Source Type

Integer
Preferred
☐

☒ Primary UID
☐ Relation UID
☒ Mandatory
☐ Deprecated

Details
Overview
UDP

Attributes:

Name

+ - < >

	Name	Data type
1	BillID	Integer
2	ResidentID	Integer
3	BillingCycle	VARCHAR (20)
4	TotalAmount	DECIMAL (2, 10)
5	PaymentStatus	VARCHAR (20)
6	DueDate	Date

Attribute Properties

Name

BillID

Data Type

☐ Domain
☒ Logical
☐ Distinct

☐ Structured
☐ Collection

Source Type

Integer
Preferred
☐

☒ Primary UID
☐ Relation UID
☒ Mandatory
☐ Deprecated

Details
Overview
UDP

Attributes:

Name

+ - < >

	Name	Data type
1	itemID	Integer
2	ItemName	VARCHAR (20)
3	CurrentStock	Integer
4	Unit	VARCHAR (10)
5	LastRestocked	Date

Attribute Properties

Name

itemID

Data Type

☐ Domain
☒ Logical
☐ Distinct

☐ Structured
☐ Collection

Source Type

Integer
Preferred
☐

☒ Primary UID
☐ Relation UID
☒ Mandatory
☐ Deprecated

DetailsOverviewUDP

Attributes:

Name

XY

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	Name	Data type
1	FeedBackID	Integer
2	ResidentID	Integer
3	Rating	Integer
4	comments	TEXT
5	SubmittedAt	Timestamp ()

Attribute Properties

Name

FeedBackID

Data Type

☐ Domain

☒ Logical

☐ Distinct

☐ Structured

☐ Collection

Source Type

Integer

Preferred ☐

☒ Primary UID

☐ Relation UID

☒ Mandatory

☐ Deprecated

DetailsOverviewUDP

Attributes:

Name

XY

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	Name	Data type
1	HotelID	Integer
2	HotelName	VARCHAR (20)
3	Capacity	Integer

Attribute Properties

Name

HotelID

Data Type

☐ Domain

☒ Logical

☐ Distinct

☐ Structured

☐ Collection

Source Type

Integer

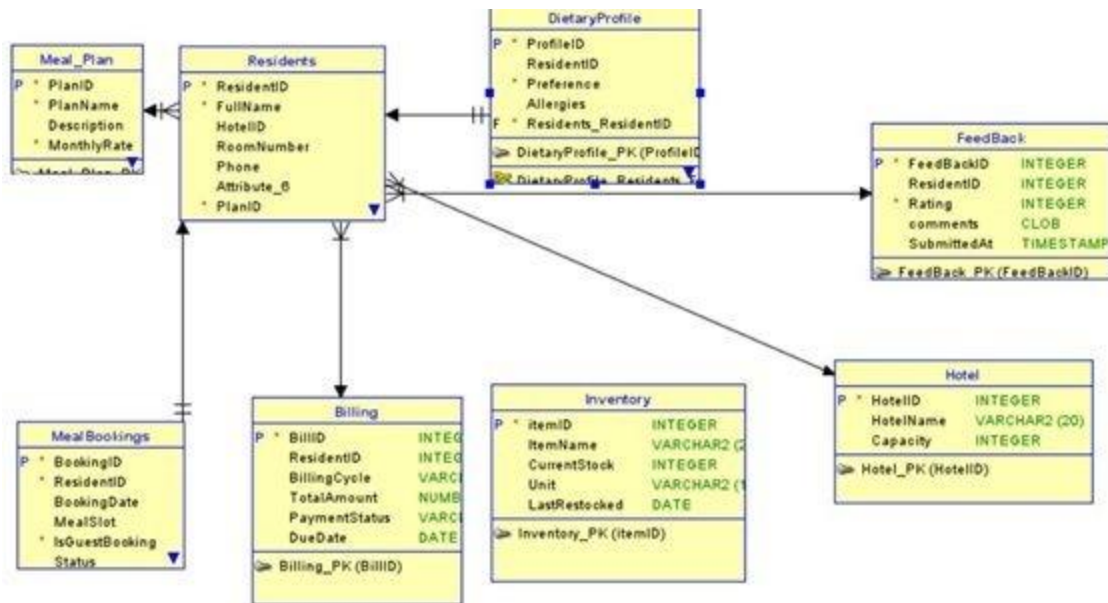
Preferred ☐

☒ Primary UID

☐ Relation UID

☒ Mandatory

☐ Deprecated



```
-- Generated by Oracle SQL Developer Data Modeler 24.3.1.351.0831
```

```
-- at:      2026-04-04 22:09:17 PKT
-- site:    Oracle Database 11g
-- type:    Oracle Database 11g
```

```
-- predefined type, no DDL - MDSYS.SDO_GEOMETRY
```

```
-- predefined type, no DDL - XMLTYPE
```

```
CREATE TABLE Billing
```

```
(
  BillID      INTEGER NOT NULL ,
  ResidentID  INTEGER ,
  BillingCycle VARCHAR2 (20) ,
  TotalAmount NUMBER (2,10) ,
  PaymentStatus VARCHAR2 (20) ,
  DueDate     DATE
)
```

```
;
```

```
ALTER TABLE Billing
```

```
ADD CONSTRAINT Billing_PK PRIMARY KEY ( BillID ) ;
```

```
CREATE TABLE DietaryProfile
```

```
(
  ProfileID      INTEGER NOT NULL ,
  ResidentID     INTEGER ,
  Preference     VARCHAR2 (20) NOT NULL ,
  Allergies      VARCHAR2 (100) ,
  Residents_ResidentID INTEGER NOT NULL
)
```

```
;
```

```
CREATE UNIQUE INDEX DietaryProfile__IDX ON DietaryProfile
```

```
(
  Residents_ResidentID ASC
)
```

```
;
```

```
ALTER TABLE DietaryProfile
```

```
ADD CONSTRAINT DietaryProfile_PK PRIMARY KEY ( ProfileID ) ;
```

```
CREATE TABLE FeedBack
```

```
(  
    FeedBackID INTEGER NOT NULL ,  
    ResidentID INTEGER ,  
    Rating      INTEGER NOT NULL ,  
    comments    CLOB ,  
    SubmittedAt TIMESTAMP  
)
```

```
;
```

```
ALTER TABLE FeedBack
```

```
ADD CONSTRAINT FeedBack_PK PRIMARY KEY ( FeedBackID ) ;
```

```
CREATE TABLE Hotel
```

```
(  
    HotelID   INTEGER NOT NULL ,  
    HotelName VARCHAR2 (20) ,  
    Capacity  INTEGER  
)
```

```
;
```

```
ALTER TABLE Hotel
```

```
ADD CONSTRAINT Hotel_PK PRIMARY KEY ( HotelID ) ;
```

```
CREATE TABLE Inventory
```

```
(  
    itemID      INTEGER NOT NULL ,  
    ItemName     VARCHAR2 (20) ,  
    CurrentStock INTEGER ,  
    Unit         VARCHAR2 (10) ,  
    LastRestocked DATE  
)
```

```
;
```

```
ALTER TABLE Inventory
```

```
ADD CONSTRAINT Inventory_PK PRIMARY KEY ( itemID ) ;
```



```

3 CREATE TABLE Meal_Plan
(
    PlanID      INTEGER NOT NULL ,
    PlanName    VARCHAR2 (20) NOT NULL ,
    Description VARCHAR2 (100) ,
    MonthlyRate NUMBER (2,10) NOT NULL
)
;

ALTER TABLE Meal_Plan
    ADD CONSTRAINT Meal_Plan_PK PRIMARY KEY ( PlanID ) ;

3 CREATE TABLE MealBookings
(
    BookingID      INTEGER NOT NULL ,
    ResidentID     INTEGER NOT NULL ,
    BookingDate    DATE ,
    MealSlot       VARCHAR2 (20) ,
    IsGuestBooking CHAR (1) NOT NULL ,
    Status         VARCHAR2 (20) ,
    Residents_ResidentID INTEGER NOT NULL
)
;

3 CREATE UNIQUE INDEX MealBookings__IDX ON MealBookings
(
    Residents_ResidentID ASC
)
;

ALTER TABLE MealBookings
    ADD CONSTRAINT MealBookings_PK PRIMARY KEY ( BookingID ) ;

```

```

CREATE TABLE Residents
(
    ResidentID          INTEGER NOT NULL ,
    FullName            VARCHAR2 (50) NOT NULL ,
    HotelID             INTEGER ,
    RoomNumber          INTEGER ,
    Phone               INTEGER ,
    PlanID              INTEGER NOT NULL ,
    Hotel_HotelID       INTEGER NOT NULL ,
    Meal_Plan_PlanID    INTEGER NOT NULL ,
    Billing_BillID       INTEGER NOT NULL ,
    FeedBack_FeedBackID INTEGER NOT NULL
)
;

ALTER TABLE Residents
    ADD CONSTRAINT Residents_PK PRIMARY KEY ( ResidentID ) ;

ALTER TABLE DietaryProfile
    ADD CONSTRAINT DietaryProfile_Residents_FK FOREIGN KEY
    (
        Residents_ResidentID
    )
    REFERENCES Residents
    (
        ResidentID
    )
;

ALTER TABLE MealBookings
    ADD CONSTRAINT MealBookings_Residents_FK FOREIGN KEY
    (
        Residents_ResidentID
    )
    REFERENCES Residents
    (
        ResidentID
    )
;

```

```
ALTER TABLE Residents
  ADD CONSTRAINT Residents_Billing_FK FOREIGN KEY
  (
    Billing_BillID
  )
  REFERENCES Billing
  (
    BillID
  )
;
```

```
ALTER TABLE Residents
  ADD CONSTRAINT Residents_FeedBack_FK FOREIGN KEY
  (
    FeedBack_FeedBackID
  )
  REFERENCES FeedBack
  (
    FeedBackID
  )
;
```

```
ALTER TABLE Residents
  ADD CONSTRAINT Residents_Hotel_FK FOREIGN KEY
  (
    Hotel_HotelID
  )
  REFERENCES Hotel
  (
    HotelID
  )
;
```

```
ALTER TABLE Residents
  ADD CONSTRAINT Residents_Meal_Plan_FK FOREIGN KEY
  (
    Meal_Plan_PlanID
  )
  REFERENCES Meal_Plan
  (
    PlanID
  )
;
```